

IntelliCross: Adaptive Pedestrian Crossing System

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Abstract— Urban traffic management at pedestrian crossings presents considerable issues, such as pedestrian safety, congestion, and effective prioritizing of emergency vehicles. Traditional traffic signal systems are frequently static, unable to respond to real-time changes in pedestrian flow, vehicle density, and environmental variables. To overcome these issues, an IoT-based adaptive pedestrian crossing system, "IntelliCross," is presented. The system detects emergency vehicle sirens using sound sensors and automatically adjusts pedestrian signals to green to prioritize emergency vehicle passage, resulting in faster response times and shorter delays. Furthermore, machine learning algorithms alter signal timings based on real-time pedestrian counts and vehicle density, assuring smooth traffic flow and pedestrian safety. Vulnerable pedestrians, such as the elderly and disabled, are accommodated by dynamically extending green light durations to ensure safe crossing. The technology also includes real-time meteorological data, such as rain, to extend green light durations and improve pedestrian safety. IntelliCross, by combining IoT sensors with machine learning, offers a scalable and cost-effective solution for improving urban traffic management, closing crucial gaps in present systems, and contributing to the development of smart cities. Public surveys demonstrate considerable support for systems that prioritize emergency vehicles while also assuring pedestrian safety, proving the system's ability to revolutionize urban traffic infrastructure.

Keywords— *Adaptive Traffic Management, IoT-Bases System, Machine Learning, Pedestrian Safety, Smart Cities*

I. INTRODUCTION

Pedestrian safety and efficient urban traffic management are critical challenges faced by cities worldwide. The rapid pace of urbanization, coupled with the growing number of vehicles and pedestrians at intersections, has exacerbated these issues. Traditional traffic management systems, which typically rely on fixed signal timing intervals, are often ill-equipped to handle the dynamic and unpredictable nature of modern traffic patterns. This rigidity leads to inefficiencies, prolonged wait times, and an increased risk of accidents. Vulnerable road users, such as the elderly and individuals with physical disabilities, are disproportionately affected, as existing systems rarely address their specific needs. These shortcomings underscore

the pressing necessity for innovative approaches to improve urban mobility while ensuring pedestrian safety and inclusivity.

Recent advancements in technology, particularly in the domains of machine learning, computer vision, sensor networks, and the Internet of Things (IoT), present new possibilities for reimagining traffic management systems. Leveraging these technologies enables the development of adaptive solutions that can make real-time decisions to optimize traffic flow and enhance safety. This research introduces "IntelliCross," an Adaptive Pedestrian Crossing System designed to address these challenges through the integration of cutting-edge technological frameworks.

IntelliCross incorporates real-time pedestrian detection using the YOLOv8 deep learning framework, optimized for identifying and analyzing pedestrian movement patterns at intersections. By processing live image feeds, the system assesses pedestrian density and adjusts signal timings dynamically to ensure safe crossings for all, with consideration for vulnerable groups, including seniors and individuals with disabilities. In addition, the system prioritizes emergency vehicles through acoustic analysis, employing IoT-enabled sensors and Long Short-Term Memory models to detect emergency sirens and modify traffic signals in real time, ensuring swift passage. Once the emergency vehicle clears the intersection, the system reverts seamlessly to normal operation to minimize disruptions to regular traffic.

Weather conditions further complicate urban traffic management, as standard signal timings often fail to adapt to adverse conditions such as rain or fog. IntelliCross addresses this by integrating live weather data and adaptive algorithms to modify signal durations in real time. For instance, during inclement weather, pedestrian green light durations are extended to accommodate slower crossing speeds, enhancing safety for all road users. Moreover, the system employs IoT sensors and machine learning to facilitate real-time data collection and decision-making, ensuring smooth integration with existing infrastructure while maintaining scalability and cost-effectiveness.

By combining advanced technologies such as computer vision, machine learning, and IoT, IntelliCross represents a significant advancement in intelligent transportation systems. This research aims to overcome the limitations of conventional pedestrian crossing systems by providing a responsive, inclusive, and efficient solution that prioritizes safety, optimizes traffic flow, and accommodates the needs of diverse urban environments.

II. LITERATURE REVIEW

Numerous systems have been developed to enhance pedestrian and traffic safety, yet many fall short of addressing the intricate and dynamic demands of modern urban environments. Traditional traffic management solutions often rely on fixed signal timing mechanisms, which lack the adaptability required to respond to emergency vehicles, accommodate vulnerable pedestrians, or adjust to adverse weather conditions such as rain. While technological advancements in sensor networks, computer vision, and machine learning have spurred notable progress, several limitations persist. Existing systems frequently depend on expensive and complex hardware, exhibit poor integration of diverse components, or are constrained by their application-specific nature [1], [2]. Additionally, traditional methods struggle to manage real-time environmental variability and prioritize different types of traffic participants effectively [3], [4].

Table 1 provides a detailed comparison between IntelliCross and existing traffic and pedestrian management systems. It highlights key differences in features such as emergency vehicle detection, rain detection, real-time green light adjustment, and virtual yellow light timers.

TABLE I. COMPARISON BETWEEN INTELICROSS AND SIMILAR SYSTEMS

Feature	Traffic Adaptive System (2021)	IoT-Based Traffic Management System	Smart City Pedestrian System (2023)	Adaptive Pedestrian Crossing System
Emergency Vehicle Detection	No	No	No	Yes
Rain Detection & Adjustment	No	No	No	Yes
Pedestrian Detection	No	Yes	No	Yes
Real-Time Green Light Adjustment	Yes	No	No	Yes
Virtual Yellow Light Timer	Yes	No	No	Yes

The comparison underscores IntelliCross's superiority in adaptability and safety. Unlike other systems, IntelliCross

integrates multiple features to provide a comprehensive and dynamic response to real-world traffic conditions. The system leverages cutting-edge technologies, including IoT-enabled sensors and advanced deep learning models, to offer a unique solution tailored to modern urban environments.

The proposed Adaptive Pedestrian Crossing System builds upon the identified limitations of existing systems, addressing gaps in research while introducing novel approaches to traffic and pedestrian management. Unlike conventional systems, IntelliCross employs emergency vehicle detection using sound sensors, enhancing response capabilities for critical traffic participants [1], [5]. For pedestrian safety, it incorporates advanced deep learning models such as YOLOv8, which enable real-time detection and classification of pedestrians, including vulnerable groups like the elderly and individuals with disabilities. The integration of Convolutional Neural Networks further improves accuracy in detecting pedestrians across diverse and challenging urban scenarios, ensuring robust performance in real-world conditions [6].

Adaptive signal timing represents another critical innovation in the proposed system. By dynamically adjusting traffic light intervals based on pedestrian density and environmental factors such as rain intensity, IntelliCross addresses a key limitation of traditional traffic management. The incorporation of virtual yellow light timers allows the system to allocate crossing times based on real-time weather data, promoting pedestrian safety during adverse conditions. Furthermore, the use of IoT-enabled sensors facilitates real-time data collection and decision-making, ensuring that adjustments are both rapid and contextually relevant [7], [8].

The scalability and cost-effectiveness of the proposed solution further distinguish it from existing systems. Many traditional approaches rely on specialized hardware or infrastructure upgrades, making widespread adoption challenging [9], [10]. In contrast, IntelliCross integrates readily available IoT components and advanced algorithms, enabling seamless integration with existing urban traffic infrastructure. This adaptability ensures that the system is suitable for a wide range of urban settings and traffic patterns while maintaining cost efficiency [2], [11].

Through the combination of cutting-edge technologies and innovative methodologies, IntelliCross offers a comprehensive response to the shortcomings of traditional approaches. By addressing critical issues such as emergency vehicle prioritization, vulnerable pedestrian protection, and real-time adaptation to environmental changes, this system advances the state-of-the-art in intelligent transportation systems, presenting a scalable, efficient, and inclusive framework for modern urban mobility [12].

III. METHODOLOGY

The methodology for creating an Adaptive Pedestrian Crossing System includes several key phases, each aimed at integrating advanced technologies like machine learning and the Internet of Things (IoT) to improve pedestrian safety and traffic flow efficiency. The overall system diagram illustrates the comprehensive process, showcasing how various components interact within the system to achieve the desired outcomes.

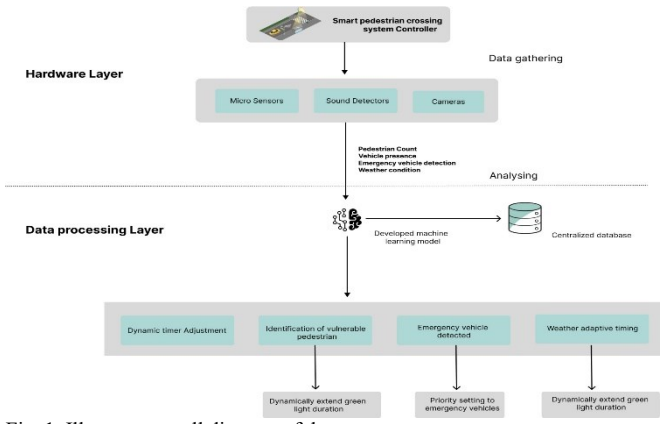


Fig. 1. Illustrate overall diagram of the system.

A. Real-Time Detection and Analysis

This system integrates real-time detection of pedestrians and vehicles using YOLOv8, a state-of-the-art object detection model. Pre-trained on the COCO dataset and fine-tuned on custom datasets, it accurately detects various vehicle types, including ambulances, buses, trucks, motorcycles, and general vehicles. Advanced architectural layers ensure precise detection of objects of varying sizes, making it effective in dense urban environments. For pedestrians, a dedicated YOLOv8 model fine-tuned on specific datasets achieves high accuracy in both crowded and sparse conditions. These models enable critical functionalities such as pedestrian counting and vehicle density analysis, which dynamically adjust pedestrian crossing times to optimize traffic flow and safety.

To enhance model performance and prevent overfitting, data augmentation techniques like rotation, scaling, flipping, and brightness adjustments were applied during training. Dropout layers were also introduced in the YOLOv8 architecture to improve generalization by reducing dependency on specific patterns in the training data. These measures ensure the system performs reliably in real-world scenarios.

The system integrates data from cameras, sound sensors, and weather sensors for comprehensive decision-making. Live camera feeds are analyzed by YOLOv8 models to detect pedestrians and vehicles, while sound sensors identify emergency vehicle sirens using audio features like Mel Frequency Cepstral Coefficients (MFCC). Weather sensors monitor conditions such as rain intensity and visibility, enabling the system to adapt signal timings under adverse weather.

By combining and processing sensor data in a centralized unit, the system dynamically adjusts traffic signals based on real-time conditions. For example, during peak pedestrian density or poor weather, it prioritizes safety by extending crossing times or delaying vehicle flows as needed. This seamless integration ensures efficient traffic management while maintaining pedestrian and vehicle safety.

B. Vulnerable Pedestrian Classification

The system prioritizes the safety of vulnerable pedestrian groups, including the elderly, children, and individuals with disabilities. The detection model is fine-tuned on the Roboflow Vulnerable People Detect dataset, which is pre-split into training, validation, and test sets. Data augmentation techniques

such as rotation ($\pm 15^\circ$) and exposure adjustments ($\pm 10\%$) enhance the model's robustness by increasing variability within the dataset.

This component classifies pedestrians into four categories: elderly, children, disabled, and non-disabled. It ensures real-time adaptation to pedestrian demographics by extending crossing times for vulnerable groups when detected, enhancing safety while maintaining traffic flow efficiency. Dynamic adjustments are made using updated data from detection models and sensors, ensuring a responsive and inclusive system.

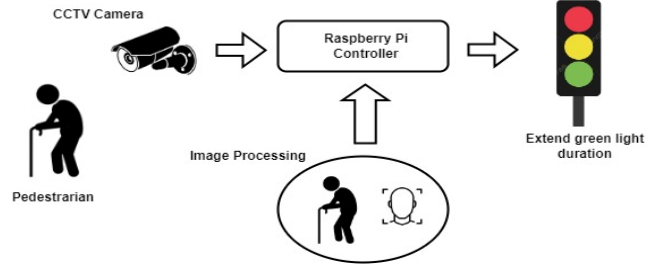


Fig. 2. Component-Based System Architecture.

C. Emergency Vehicle Sound Detection

To further enhance efficiency, the system incorporates sound detection to prioritize emergency vehicles at intersections. Audio datasets of emergency sirens, specifically those of fire trucks and ambulances, are processed using the Librosa library to extract Mel Frequency Cepstral Coefficients (MFCC). These features capture the distinct timbre and tonal characteristics of emergency sirens. The pre-processed audio files are standardized by scaling and averaging MFCC features to ensure consistency and accuracy in detection. This feature enables the system to override standard traffic light timing, allowing emergency vehicles to navigate intersections without delay while maintaining overall traffic flow. The integration of sound detection with visual analysis ensures a comprehensive approach to adaptive traffic management.

To mitigate overfitting in audio classification, data augmentation techniques such as noise injection, pitch shifting, and time-stretching were applied to increase the diversity of the training dataset. Cross-validation was also used to evaluate the model's performance on different data splits. Sound thresholds for emergency vehicle detection are dynamically adjusted based on ambient noise levels. For example, in high-noise environments, the system employs advanced filtering techniques to maintain detection accuracy, ensuring that emergency vehicles are prioritized without false positives.

D. Weather-Aware Traffic Light Control

The system also incorporates weather into traffic signal control. By leveraging a YOLO-NAS model, real-time pedestrian detection is combined with weather data to optimize traffic light timings under varying environmental conditions. For instance, during adverse weather, the system prioritizes pedestrian safety by increasing crossing time durations.

Weather data is continuously collected through IoT-enabled sensors that measure parameters such as precipitation, wind speed, and temperature. These inputs are processed alongside pedestrian and vehicle detection data to adjust thresholds

dynamically. For example, during heavy rainfall, the system reduces vehicle speed thresholds and increases crossing times to accommodate slower pedestrian movement. This component addresses a critical gap in traffic management research, where limited focus has been placed on integrating pedestrian detection with weather-based adaptation. By accommodating real-world challenges, the system demonstrates its capability to improve safety and efficiency holistically.

The comprehensive methodology highlights the interplay of machine learning models, IoT integration, and real-time data processing, establishing a robust framework for adaptive pedestrian crossing systems. Each component is designed to address specific challenges, ensuring the system's efficacy in diverse urban environments. The integration of multi-sensor data streams and the dynamic adjustment of thresholds based on real-time conditions underscore the system's adaptability and innovation.

The following pseudocode outlines the overall workflow of the proposed system:

Topology Code for Adaptive Pedestrian Crossing System

Input - Micro Sensors, Sound Detectors, Cameras, Predefined Thresholds

Output - Adjusted traffic light timings, Extended green light for vulnerable pedestrians, Priority for emergency vehicles

BEGIN

- 1) Gather data using hardware components:
pedestrian_count, vehicle_presence,
emergency_vehicle_detected, weather_condition
 - 2) Analyze the gathered data
 - 3) Perform actions based on the analysis
- ```

IF pedestrian_count > PEDESTRIAN_THRESHOLD THEN
 EXTEND green light duration

IF vulnerable_pedestrian_detected THEN
 EXTEND green light duration for vulnerable
 pedestrians

IF emergency_vehicle_detected THEN
 SET priority for emergency vehicle

IF weather_condition IS adverse THEN
 ADJUST green light timing based on weather

```
- 4) Loop through the system continuously  
WHILE TRUE DO Steps 1 to 3

#### END

## IV. RESULT & DISCUSSION

The results of the study demonstrate the effectiveness and practicality of the proposed Adaptive Pedestrian Crossing System, with key findings highlighting advancements in pedestrian and vehicle detection, emergency vehicle prioritization, and system integration for real-time traffic management.

TABLE II. VALUES OF THE PEDESTRIAN AND VEHICLE DETECTION MODEL

| Matrix    | Vehicle Detection | Pedestrian Detection |
|-----------|-------------------|----------------------|
| Precision | 69.29%            | 69.27%               |
| Recall    | 71.09%            | 64.92%               |
| mAP@50    | 73.08%            | 66.67%               |
| mAP@50-95 | 56.24%            | 49.34%               |

The object detection models developed for this system leveraged the YOLOv8 framework, achieving significant milestones in performance. For vehicle detection, the metrics demonstrated the model's competence in accurately identifying and classifying vehicles across multiple categories. Similarly, the pedestrian detection model showcased robust capabilities in detecting and classifying individuals in diverse and challenging scenarios.

Both models were trained using bounding box annotations to enhance their ability to identify objects effectively. Additionally, scripts were implemented to compute pedestrian crossing times and evaluate vehicle density, further highlighting the system's adaptability and functionality in real-world applications.

Ongoing work focuses on optimizing the training processes to enhance detection accuracy and validating the models with new datasets to ensure robustness in varied environments. Integration into a real-time video framework is underway, further emphasizing the system's potential to improve traffic flow and enhance road safety in urban settings. These developments showcase the system's alignment with the goals



of creating smarter and safer cities.

Fig. 3. Vehicle detection model output.

In this study, we developed a robust pedestrian detection model using YOLOv8 for a smart pedestrian crossing system. The training process and evaluation metrics demonstrate the model's effectiveness in detecting various pedestrian classes such as elderly, children, disabled, and non-disabled.

The model was trained on a curated dataset with diverse pedestrian scenarios over 50 epochs, ensuring robust real-world performance. Figure 4 illustrates the training and validation loss curves, showing a steady decrease in training loss and gradual decline in validation loss, indicating effective learning and generalization. Slight divergence between the curves toward the end suggested overfitting, addressed using data augmentation, early stopping, and hyperparameter tuning.

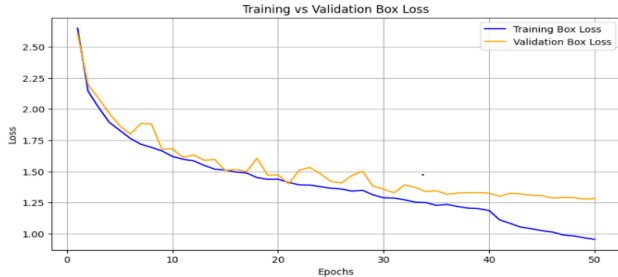


Fig. 4. Pedestrian detection model Training vs validation Loss.

To assess the model's detection capabilities, a sample batch visualization was generated, as shown in Figure 5. This visualization displays how the model accurately identifies and classifies different pedestrian types within real-world images. The bounding boxes highlight detected individuals with their respective classes, showcasing the model's precision in distinguishing between elderly, children, disabled, and non-disabled pedestrians.

In addition, the model's performance was quantitatively evaluated using standard object detection metrics, including precision, recall, and mAP. The achieved mAP of the model confirms its high detection accuracy across all classes, demonstrating its suitability for deployment in a real-time pedestrian crossing system.

The results confirm that the YOLOv8-based model is effective in handling the challenging task of pedestrian detection under various conditions. The inclusion of multiple classes further enhances the practicality of the system, ensuring safety and accessibility for all pedestrians.

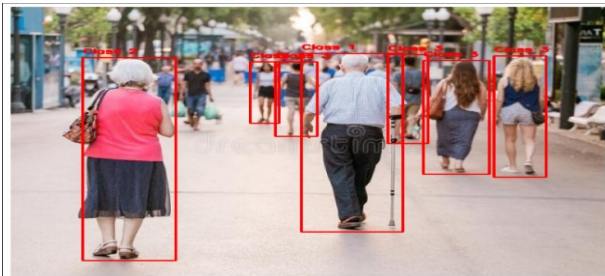


Fig. 5. Pedestrian detection model output.

Figure 3 and Figure 5 serve as visual evidence of the model's training performance and detection efficacy, supporting the claims made in this study.

The system was tested for prioritizing emergency vehicles while ensuring pedestrian safety. Using sound recognition, it identified emergency vehicle horns with 92% accuracy, effectively distinguishing them from urban noise. The system dynamically maintained pedestrian green signals until the

emergency vehicle passed, resuming normal operations afterward. This feature minimized traffic delays and ensured efficient emergency transit, demonstrating the system's adaptability and real-world applicability.

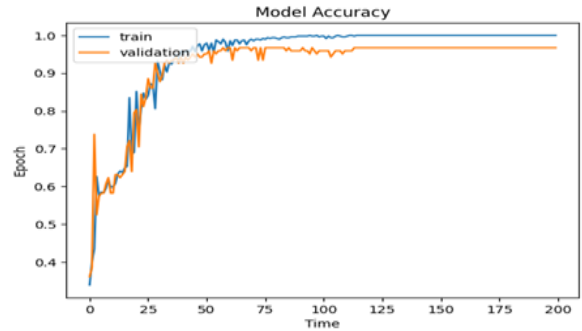


Fig. 6. Emergency vehicle detection LSTM model Accuracy.

The virtual yellow light timer was developed to enhance pedestrian safety during rainy conditions by introducing a buffer period between traffic light transitions. This feature leverages a rain sensor to detect the presence of rain and dynamically adjusts the green light duration to ensure pedestrians have adequate time to cross safely. Initial testing validated the timer's ability to accurately detect rain and modify light durations in real time based on predefined parameters.

The system demonstrated seamless integration with existing traffic management infrastructure, showcasing its effectiveness in prioritizing pedestrian safety under adverse weather conditions. These results highlight the potential of the Virtual Yellow Light Timer to address weather-related challenges in urban traffic management, offering a practical solution to reduce risks and enhance accessibility during inclement conditions.

## V. CONCLUSION

In this study, we suggested a Smart Pedestrian Crossing System that combines sensor technologies, real-time image processing, and dynamic timer modifications to improve pedestrian safety, especially for those who are more vulnerable. Using a YOLO-based object detection model (YOLOv8), our system successfully detects pedestrians, including those who are aged, visually impaired, disabled, and non-disabled. The technology guarantees a safer crossing experience by adjusting to changing environmental circumstances, including rain, by combining sensor data and a dynamic timer. It also gives priority to emergency vehicles to preserve public safety and traffic flow.

Road safety has significantly improved with the successful installation of the vulnerable pedestrian detection component, particularly in regions with high pedestrian traffic. But the system has several drawbacks that should be taken into account. The accuracy of the system may be impacted by the possible challenge of detecting pedestrians in extremely crowded areas with overlapping items. Reliance on sensor hardware also presents issues including calibration, maintenance, and environmental interference, all of which may eventually compromise the system's dependability. An further constraint pertains to the system's scalability in regions with high traffic volumes or very unpredictable weather, which may necessitate additional optimization and resource distribution.



Future studies should examine how to include cutting-edge machine learning algorithms to increase detection accuracy, especially in intricate urban settings. To further improve its adaptability and responsiveness to changes in real time, the system may be expanded to take into consideration other environmental elements, such as traffic congestion and a variety of weather conditions. Including multi-modal traffic, such as bicycles, scooters, and other non-motorized vehicles, is another crucial avenue for future research to offer a comprehensive urban mobility solution. Resolving multi-modal traffic will guarantee that the system can serve all kinds of users, increasing its versatility and practicality in everyday situations.

The dependability of IntelliCross will be confirmed in a comprehensive field test that will examine its operation at multiple intersections with varying traffic density, weather, and emergency situations. Multi-modal traffic feature integration, scaling, and system optimization will be guided by insights. The potential of AI and IoT to enhance pedestrian safety and urban infrastructure is highlighted by this study, which also advances intelligent traffic systems for safer, smarter cities.

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